Improving Education Access and Consistency with AI Driven Facilitation

Project Scope, Objectives and Literature Review

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Disclaimer *– All these conference papers have been submitted as partial fulfilment for the*

project requirement for the BE(Hons) degree. Although they have been assessed, no errors or factual information have been corrected or checked.

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1. INTRODUCTION

Access to quality education remains a significant global challenge, particularly in underserved and resource-limited environments. Many communities continue to face limited access to qualified educators, inconsistent delivery of learning, and barriers to establishing sustainable educational programmes. These challenges affect not only access to education but also the quality and reliability of learning outcomes.

In reply to these challenges, the present project investigates the design of a facilitator-based education platform to strengthen accessibility, affordability, and reliability in educational delivery. The core concept involves separating educational expertise from classroom delivery: lessons are designed by subject-matter experts and delivered by facilitators with structured, step-by-step guidance. This model aims to reduce dependence on highly trained teachers while assuring consistent lesson delivery.

Artificial intelligence (AI) is incorporated into this model as a supporting tool to enhance content generation, personalisation, and resource creation. The primary focus of the project is not on AI in isolation, but rather on the effective integration of AI within structured learning systems that support younger learners.

This report outlines the project scope and objectives and provides a literature review examining how current AI-based and traditional education systems address these challenges, as well as how the proposed model addresses the identified gaps.

2. PROJECT SCOPE & RESEARCH OBJECTIVES

2.1 Problem Statement

Despite advances in educational technology, significant problems persist in delivering consistent, high-quality education at scale. Current systems remain heavily dependent on qualified teachers for both lesson design and delivery, creating a structural bottleneck that limits scalability and contributes to uneven learning quality, particularly in underserved or resource-constrained regions. While AI-supported and online platforms offer greater flexibility, they typically presuppose a level of learner independence that is unsuitable for younger students, who require structured guidance, supervision, and interactive support. Furthermore, access to AI-enhanced content does not guarantee consistent or effective instructional delivery; variability in teaching approaches, the absence of lesson structures, and limited contextual adaptability continue to produce inequitable learning outcomes. These compounding limitations point to a clear gap in existing provision: the absence of a scalable system that decouples expert lesson design from classroom delivery and embeds structured support throughout the learning process.

2.2 Research Aims and Question

This research aims to investigate the role of artificial intelligence in supporting learning for K–12 students and to identify the limitations that affect its effectiveness for younger learners. The central research question guiding this study is:

> **How is artificial intelligence currently used to support learning for K–12 students, and what limitations affect its effectiveness for younger learners?**

To address this question, the research seeks to analyse existing applications of AI in education, evaluate their benefits and limitations, and explore how these findings can inform the design of more effective educational systems.

2.3 General Approach

This project employs a research-informed design methodology that combines literature analysis with system design principles. The study begins with a comprehensive review of research on artificial intelligence in education and current educational delivery models, focusing on their benefits, limitations, and relevance to K–12 contexts. Insights from the literature are synthesised to identify key gaps, particularly regarding structured learning, accessibility, and scalability. Based on these findings, a conceptual facilitator-based platform is proposed to separate lesson creation from classroom delivery while integrating AI as a supporting tool for content generation and learning enhancement.

The proposed system is evaluated conceptually by assessing its alignment with the identified gaps and its potential to improve consistency, accessibility, and overall learning effectiveness.

2.4 Objectives

The project aims to:

- Investigate challenges related to access, quality, and teacher dependency in education.
- Analyse current AI-based and traditional learning systems
- Identify limitations in self-directed and technology-supported learning.
- Design a platform model that separates lesson design from classroom delivery.
- Develop a structured facilitator-based learning approach.
- Incorporate AI features to support content generation and learning.
- Demonstrate the feasibility of the proposed system design.
- Evaluate the solution against the gaps currently identified in K-12 education.

2.5 Project Scope

The scope of this project is restricted to the design and development of a proof-of-concept platform for facilitator-based lesson delivery. The primary focus is on establishing the essential system framework and capabilities required to support guided classroom instruction, rather than developing a comprehensive or fully deployable educational system.

2.6 Value of the Project

This project addresses key limitations present in both traditional and AI-supported education systems.

By combining structured lesson delivery with facilitator-based teaching and AI support, the platform aims to:

- Improve consistency in educational delivery.
- Reduce dependence on highly trained teachers.
- Increase accessibility to quality education.
- Enable scalable learning environments.

The proposed system provides a balanced approach by integrating technology with guided instruction, thereby supporting more effective learning experiences for K–12 students.

3. LITERATURE REVIEW

3.1. Overview of Artificial Intelligence in Modern Education

The rapid expansion of artificial intelligence (AI) in education is closely associated with the proliferation of digital and online learning environments. As online platforms became more prevalent, educational delivery increasingly

relied on technology, with AI tools integrated to support content delivery, interaction, and assessment *(Kasneci et al., 2023; Alshahrani, 2023).* This shift represents *a greater movement* away from traditional classroom-based instruction toward technology-driven, flexible learning environments, enabling students to engage with digital platforms independently, manage their learning pace, and access resources without continuous teacher supervision *(Lee & Zhai, 2024)*. Concurrently, the educator's role transitioned from direct instruction to facilitation and support *(Touretzky et al., 2019), fostering student-centred learning* with greater emphasis on autonomy and personalised experiences.

Figure 3.1 illustrates that artificial intelligence in education can be conceptualised in three progressive stages: function, utilisation, and impact. This framework illustrates how AI develops from basic technical capabilities to ultimately influencing learning outcomes (Lang, 2025), providing a foundation for examining specific methods of AI integration within educational *environments*.

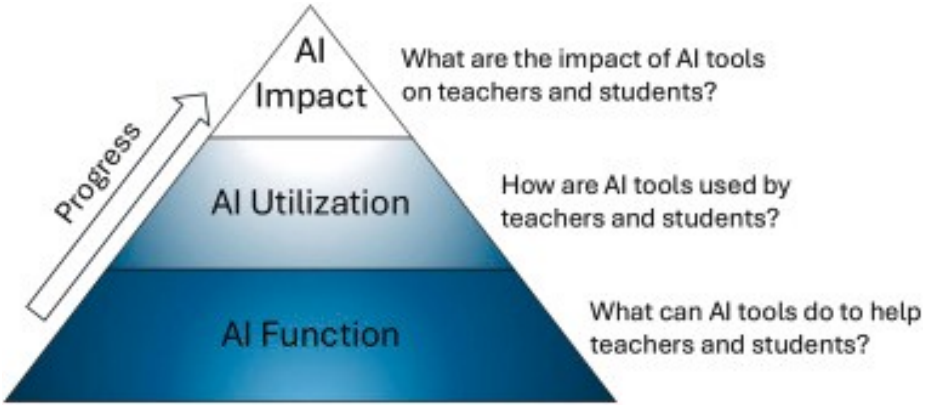


Figure 3.1: Progressive Stages of Artificial Intelligence Application in Education (sourced from Lang, 2025)

Despite these advancements, the transition to AI-enabled platforms has not benefited all learners equally. While such platforms have improved educational access, they have also introduced variability in learning experiences. Some students adapted effectively to independent environments, whereas others struggled without structured guidance *(Vilcarino & Langreo, 2025)*. Additionally, the rapid adoption of AI has surpassed the development of effective implementation frameworks, resulting in inconsistent application across educational contexts (Lang, 2025). This tension between flexibility and consistency, along with the absence of standardised guidance, illustrates the need to better understand how AI supports learning, particularly for younger students who require structured and guided environments.

3.2. Applications and Benefits of AI in Younger Education (K-12)

AI has been implemented in education through personalisation of learning, task automation, and support for teaching processes. A prominent application is adaptive learning systems, which adjust content based on student performance, allowing learners to access materials tailored to their needs and progress at an individual pace *(Lang, 2025; Pardamean et al., 2022)*. For example, a study with 269 primary school students demonstrated that AI-based personalised learning significantly improved student performance, showing the effectiveness of adaptive approaches *(Pardamean et al., 2022)*. Furthermore, AI systems deliver immediate feedback, enabling students to identify errors and enhance understanding in real time *(Samala et al., 2024).* This continuous feedback loop promotes active learning and facilitates rapid error correction, particularly in practice-intensive subjects such as mathematics and language learning, thereby increasing efficiency compared to traditional methods where feedback is delayed.

AI has also contributed to enhanced accessibility and efficiency in education. AI-based platforms provide students with the ability to access learning materials at any time and from any location, increasing flexibility and inclusivity, especially for those in remote or underserved regions where traditional resources are scarce (Rane, 2024). Additionally, AI adapts content to accommodate various learning styles and abilities, and automates administrative tasks such as grading, content generation, and data analysis *(Pisica et al., 2023)*. These efficiencies reduce educators' administrative workload, enabling greater focus on student interaction and support, and promote scalability and resource optimisation in educational settings.

Despite these advantages, research on AI's effectiveness in education has produced mixed results. An umbrella review of 102 studies indicated that although AI enhanced engagement and personalisation, its success was highly dependent on implementation quality and student interaction *(Lang, 2025)*. While some studies reported notable improvements in learning outcomes, others observed minimal impact, indicating that AI alone does not ensure improved learning and must be thoughtfully integrated into educational processes. This issue is particularly evident in the context of self-directed learning, where AI's effectiveness is closely linked to learners' skills and dispositions.

3.3. AI in Self Directed Learning (SDL)

Self-directed learning (SDL) is an educational approach in which students assume responsibility for planning, monitoring, and evaluating their own learning. Research indicates that AI tools can facilitate SDL by offering personalised guidance, feedback, and adaptive learning pathways, *thereby encouraging* students to take a more active role and enhancing motivation and engagement (Ng et al., 2024; Achuthan, 2025). A meta-analysis of 32 studies with 3,029 participants found that AI-based interventions significantly improved both self-regulated and self-directed learning *(Achuthan, 2025). Similarly, a study involving 74 students over three weeks reported improvements in* engagement, motivation, and learning outcomes *(Ng et al., 2024)*.

As illustrated in Figure 3.2, AI interventions support different dimensions of self-regulated learning, including cognitive, motivational, and behavioural processes, which contribute to the development of self-directed learning *(Achuthan, 2025)*.

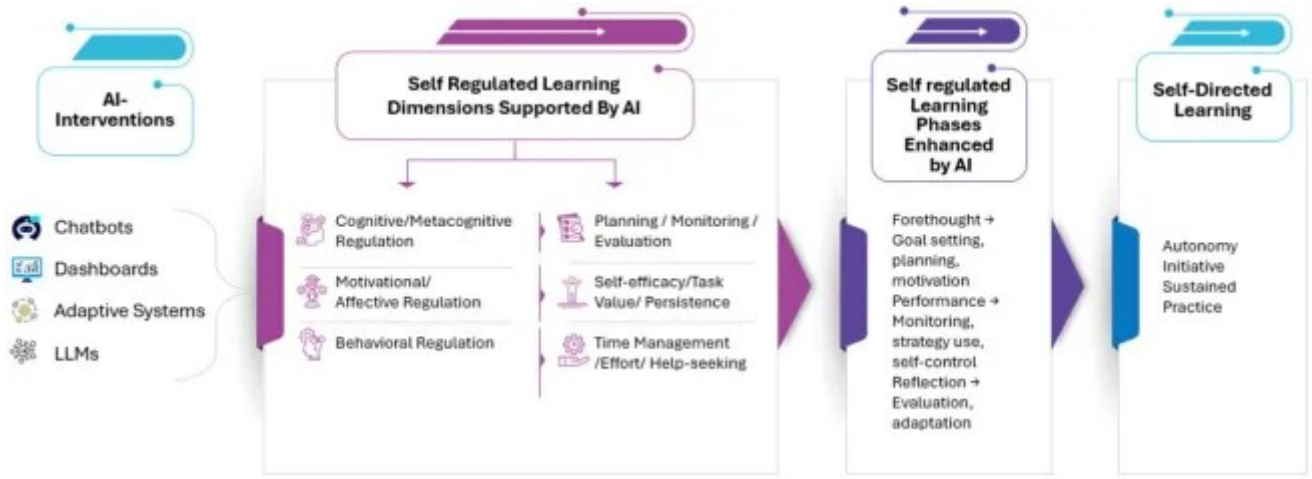


Figure 3.2: AI–Regulation–Autonomy Pathway in Learning (sourced from Achuthan, 2025)

This demonstrates AI's function as a scaffold that assists learners in planning, monitoring, and evaluating their progress. However, its effectiveness is contingent upon the learner's ability to engage independently with these processes. Research consistently shows that students need strong self-regulation skills, such as time management, discipline, and progress monitoring, to use AI tools effectively (Shen & Teng, 2024). Earlier studies also found that students in online environments often struggled with independent learning *(Chen, 2009; Kim et al., 2014). Without these foundational skills, students not only failed to benefit from AI's features but also risked developing over-reliance, which is associated with diminished critical thinking and problem-solving abilities (Kasneci et al., 2023)*.

Collectively, these findings suggest that AI is most effective when learners possess strong self-directed learning skills, a condition less common among younger students in K–12 settings. This raises significant concerns regarding the suitability of AI-driven approaches for this age group, which are examined in the subsequent section.

3.4. Limitations of AI for Younger Learners (K-12)

Despite its potential, AI exhibits several limitations when implemented with younger learners in K–12 education. A primary challenge is the presumption of self-directed learning; younger students frequently struggle to manage their learning independently, resulting in reduced engagement and inconsistent outcomes *(Chen, 2009)*. Numerous studies

have found that students often rely excessively on AI tools, using them to complete tasks without fully understanding the material and, in some cases, bypassing essential learning processes (NPR, 2026). A review of 22 studies identified additional risks, such as cognitive dependency, diminished creativity, and negative effects on student well-being (Tao et al., 2026), *raising broader concerns that* AI may supplant rather than support learning.

Researchers have also observed that AI systems are unable to provide the structured guidance and meaningful human interaction essential for younger learners *(Healey, 2025)*. Unlike teachers, AI cannot adequately address students' emotional and behavioural needs, which may decrease engagement and motivation, particularly among younger students. Additional studies have identified psychological and developmental concerns, such as reduced confidence, social isolation, and increased dependence on technology *(Tao et al., 2026), which are particularly significant for learners still developing foundational cognitive and social skills. Broader systemic issues,* including data privacy, ethical considerations, and unequal access to technology, further contribute to inconsistent implementation and variable learning experiences across educational contexts (Lang, 2025).

In summary, these findings demonstrate that although AI presents potential benefits in K–12 education, it does not fully address the developmental and pedagogical requirements of younger learners.

3.5. Implications for AI-Supported Learning System Design

The literature suggests that effective integration of AI in education necessitates careful system design that considers both its advantages and limitations. Researchers consistently emphasise that AI should serve as a complement to, rather than a replacement for, teachers, supporting the delivery of structured and meaningful learning experiences *(Kasneci et al., 2023)*.

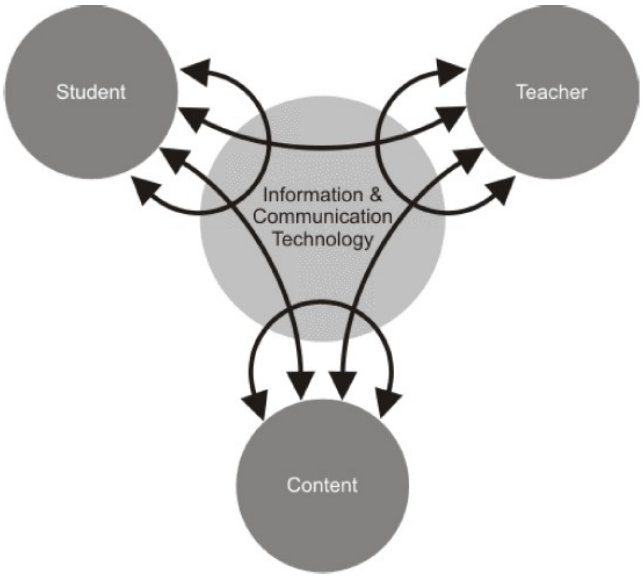


Figure 3.3: Interaction Between Students, Teachers, and Learning Content in AI-Supported Learning Environments (sourced from Pardamean et al., 2022)

Figure 3.3 demonstrates that AI systems facilitate interactions among students, teachers, and learning content, fostering more connected and adaptive learning environments (Pardamean et al., 2022). This underscores the importance of integrating AI within structured systems that guide students through learning processes. Research highlights the value of guided learning environments, where students follow clear steps and receive consistent support; structured lesson flows help maintain engagement and minimise confusion or disconnection (Lang, 2025). Rather than relying solely on AI, educational systems should enable facilitators to guide students, monitor progress, and provide additional support as needed. This hybrid approach combines the scalability of AI with the effectiveness of human interaction (Pisica et al., 2023). AI should be used selectively to enhance, rather than replace, core learning processes *(Samala et al., 2024), supporting* activities such as content generation, feedback, and practice, while educators remain responsible for ensuring meaningful engagement. Centralised content management and structured lesson design further ensure consistent educational quality for all students, regardless of location or facilitator.

Overall, the literature suggests that future educational systems should incorporate AI within structured, facilitator-supported environments, balancing the scalability and flexibility of AI with the consistency and human connection necessary for effective learning.

3.6 Identified Gaps in AI-Supported Education

Building on the preceding sections, the literature reveals several consistent gaps in the current implementation of AI within K–12 education. While AI demonstrates significant potential to enhance personalisation, accessibility, and learning efficiency, existing systems do not consistently address the pedagogical and developmental needs of younger learners, nor do they adequately account for the practical constraints of real-world delivery at scale. These gaps inform the need for more structured, scalable, and context-sensitive approaches to AI-supported education.

3.6.1 Personalisation and the Delivery Gap

A substantial body of research confirms that personalised learning improves student engagement, achievement, and motivation by adapting content to individual needs and abilities *(Pardamean et al., 2022; Samala et al., 2024)*. AI systems have demonstrated effectiveness in supporting this through adaptive content delivery and real-time feedback. However, the literature also reveals that current implementations focus predominantly on content-level personalisation, with considerably less attention given to how that learning is structured and delivered in practice. While materials may be tailored to individual students, the overall learning experience often remains inconsistent and context-dependent, pointing to a gap between personalised content generation and structured instructional delivery. Effective AI-supported learning, therefore, requires integrating both elements, not merely adapting content.

3.6.2 Dependence on Teacher Expertise and Scalability Constraints

The literature consistently highlights that current educational models remain heavily dependent on teacher expertise for both the design and delivery of learning experiences. Although AI can support content generation and feedback, its effective implementation still relies on educators to structure and facilitate learning *(Lang, 2025)*. This reliance creates significant scalability challenges, particularly in contexts where access to highly trained educators is limited or unreliable. The findings suggest that instructional design and classroom delivery are not sufficiently separated in existing systems, a bottleneck that constrains the reach and consistency of AI-supported education and that the literature does not yet offer a clear solution to.

3.6.3 Lack of Structured Guidance in AI-Supported Learning

A consistent limitation identified across studies is the assumption of self-directed learning capability in AI-based environments. As established in sections 3.3 and 3.4, younger learners frequently lack the self-regulation skills required to engage with AI tools independently and effectively *(Chen, 2009; Shen & Teng, 2024)*. Without a clear instructional structure, students are prone to confusion, disengagement, or over-reliance on AI-generated outputs, outcomes that are particularly damaging in K–12 contexts where foundational skills are still being developed. The literature, therefore identifies a gap in which flexibility is prioritised over structured support, thereby reducing the effectiveness of AI for learners who arguably need it most.

3.6.4 Accessibility and Equity Gap

Although AI has been shown to improve access to learning materials through flexible, on-demand delivery *(Rane, 2024)*, access to high-quality education remains uneven. Factors such as teacher availability, institutional infrastructure, and geographic location continue to shape educational outcomes in ways that AI alone cannot resolve *(Lang, 2025)*. Current implementations primarily enhance access to content, rather than addressing the broader challenge of delivering consistent, high-quality learning experiences across diverse contexts. This distinction between access to resources and access to effective education represents a meaningful gap in the literature, one that system design must explicitly address.

3.6.5 Limited Adaptability Across Contexts

While AI systems can personalise content at the individual level, the literature indicates that many platforms lack adaptability across broader educational contexts. Variations in curriculum, learner age, subject area, and cultural environment are not consistently accommodated, limiting the transferability of AI-supported systems across settings (Achuthan, 2025). Most studies evaluate AI tools within specific institutional or geographic contexts, making it difficult to draw generalisable conclusions about their effectiveness elsewhere. The findings, therefore, highlight a need for systems that support flexible instructional design at the platform level, enabling content and delivery to be modified to meet diverse contextual requirements rather than being fixed to a single educational environment.

3.6.6 Synthesis

Collectively, the literature reveals a consistent pattern: current AI-based educational systems emphasise content personalisation but overlook instructional delivery; remain dependent on teacher expertise in ways that limit scalability; assume levels of learner independence that are often inappropriate for younger students; improve access to content without ensuring equitable learning outcomes; and lack the adaptability required to function effectively across diverse educational contexts. Taken together, these gaps indicate that the promise of AI in education has yet to be fully realised in practice, particularly for K–12 learners in non-traditional or under-resourced settings. Addressing these limitations is essential to achieving the broader goals of accessibility, consistency, and quality in education and provides the foundation for the system design proposed in this study.

4. PROJECT SCHEDULE GANTT CHART

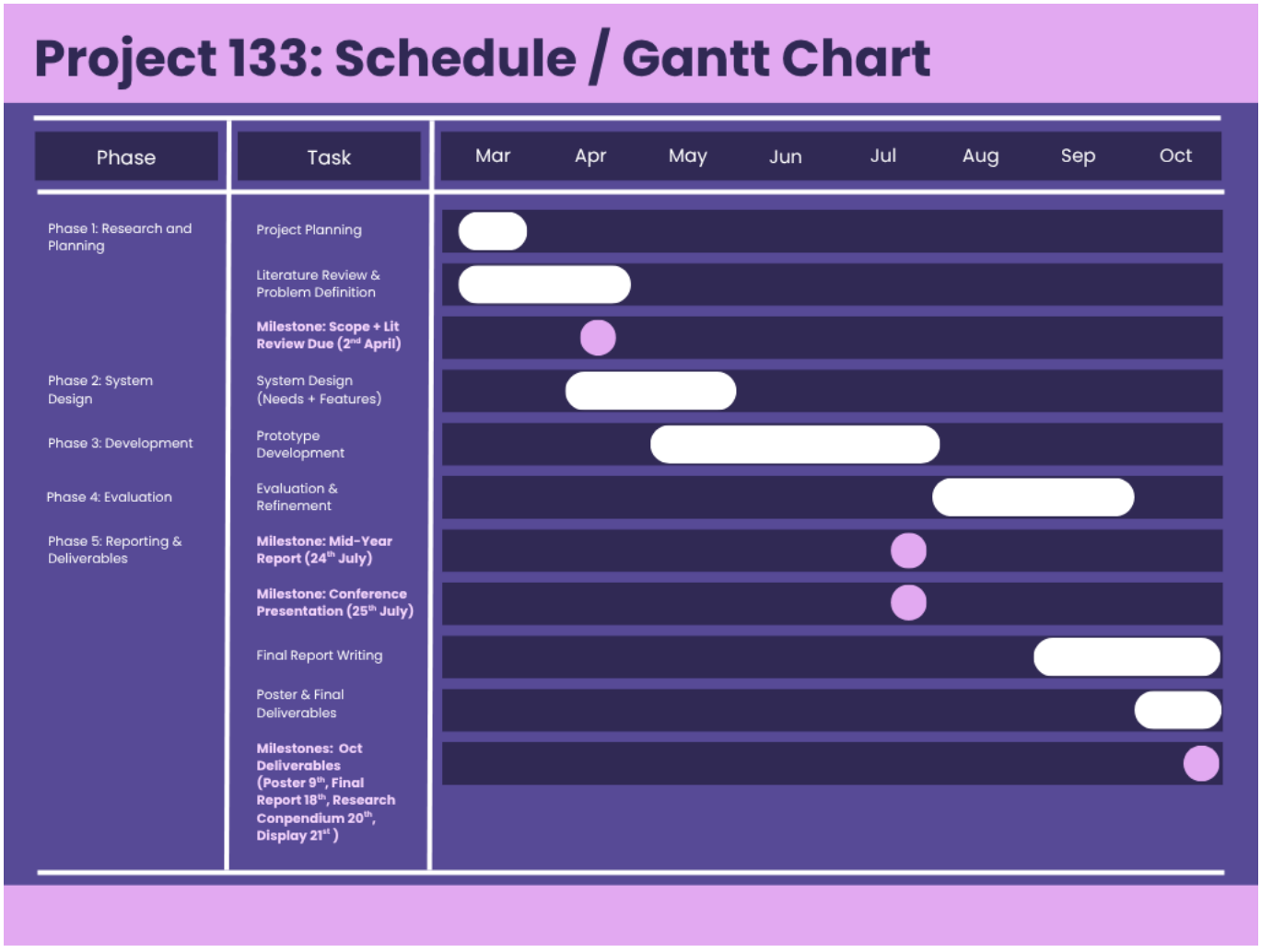


Figure 4.1: Project Timeline & Gantt Chart

The project timeline is structured across key phases, beginning with research and literature review, followed by system design, development, and evaluation. Major milestones, including the literature review submission, mid-year

report, and final deliverables, are aligned with the course schedule. This phased approach ensures steady progress and allows sufficient time for refinement and completion of all project components.

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